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Processing method and substrate processing system

Abstract

A processing method is implemented in a substrate processing system. The system includes: a processing container including a stage on which a substrate is placed; an attachment portion to which a raw material container is detachably attached; a heater configured to produce a raw material gas; a carrier gas supplier configured to supply a carrier gas to the raw material container; a supply line configured to supply the raw material gas with the carrier gas; a controller configured to control at least one of the heater and a flow rate of the carrier gas supplied from the carrier gas supplier; and a determination portion configured to determine an initial filling state of the raw material container attached to the attachment portion. The processing method includes determining the initial filling state of the raw material container based on an operation record and a table.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage Application of International Patent Application No. PCT/JP2020/020191, having an International Filing Date of May 21, 2020, which claims the benefit of priority to Japanese Patent Application No. 2019-114040, filed Jun. 19, 2019, each of which is hereby incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure relates to a processing method and a substrate processing system.

BACKGROUND

(3) There is known a substrate processing apparatus which sublimates a solid raw material at room temperature, supplies the raw material gas with a carrier gas to the processing container, and performs a desired process on a substrate in a processing container.

(4) Patent Document 1 discloses that tungsten chloride, which is a solid raw material, is sublimated to generate tungsten chloride gas. A method of forming a tungsten film is also disclosed and includes: supplying a tungsten chloride gas as a tungsten raw material and a reduction gas to a substrate simultaneously or alternately under a depressurized atmosphere; and causing the tungsten chloride gas and the reduction gas to be reacted with each other while heating the substrate, so that a tungsten film is formed on the surface of the substrate.

PRIOR ART DOCUMENTS

Patent Documents

(5) Patent Document 1: Japanese Laid-Open Patent Publication No. 2015-193908

(6) An aspect of the present disclosure provides a processing method and a substrate processing system which stably supply a raw material gas to a processing container.

SUMMARY

(7) In order to solve the above problem, an aspect of the present disclosure provides a processing method of a substrate processing system including: a processing container including a stage on which a substrate is placed; an attachment portion to which a raw material container configured to accommodate a solid raw material is detachably attached; a heater configured to produce a raw material gas by vaporizing a solid raw material accommodated in the raw material container; a carrier gas supplier configured to supply a carrier gas to the raw material container; a supply line configured to supply the raw material gas with the carrier gas from the raw material container to the processing container; a controller configured to control at least one of the heater and a flow rate of the carrier gas supplied from the carrier gas supplier in order to control a flow rate of the raw material gas which is supplied to the processing container; and a determination portion configured to determine an initial filling state of the raw material container attached to the attachment portion, wherein the processing method comprises determining, by the determination portion, the initial filling state of the raw material container based on an operation record and a table.

(8) According to an aspect of the present disclosure, it is possible to provide a processing method and a substrate processing system which stably supply a raw material gas to a processing container.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an example of a schematic cross-sectional view of a substrate processing system according to an embodiment.

(2) FIG. 2 illustrates an example of a gas supply sequence when forming a film through a CVD process.

(3) FIG. 3 illustrates an example of a gas supply sequence when forming a film by the ALD process.

(4) FIG. 4A is a schematic view illustrating an example of a raw material container.

(5) FIG. 4B is a schematic view illustrating an example of a raw material container according to a reference example.

(6) FIG. 4C is a schematic view illustrating an example of a raw material container according to the present embodiment.

(7) FIG. 4D is a schematic view illustrating an example of another raw material container according to the present embodiment.

(8) FIG. 5 is an example of a graph illustrating the relationship between the weight of a solid raw

material and the total surface area of the solid raw material.

(9) FIG. 6 is an example of a graph illustrating the relationship between the weight of a solid raw material and the total surface area of the solid raw material.

(10) FIG. 7 is an example of a graph illustrating the relationship between the weight of a solid raw material and the total surface area of the solid raw material.

DETAILED DESCRIPTION

(11) Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. In each of the drawings, the same components are denoted by the same reference numerals, and redundant descriptions thereof may be omitted.

(12) <Substrate Processing System>

(13) The substrate processing system according to the present embodiment will be described with reference to FIG. 1. FIG. 1 is an example of a schematic cross-sectional view of a substrate processing system according to the present embodiment. The substrate processing system includes a substrate processing apparatus **100**, a plurality of attachable/detachable raw material containers **32**, and a determination device **200** configured to determine an initial filling state of the raw material container **32**.

(14) The substrate processing apparatus **100** is an apparatus for forming a tungsten film on the surface of a substrate W, such as a wafer, by supplying a tungsten chloride (WCl.sub.6) gas as a film-forming raw material gas and H.sub.2 gas as a reduction gas to the substrate W. The substrate processing apparatus **100** includes, for example, a chemical vapor deposition (CVD) apparatus, an atomic layer deposition (ALD) apparatus, and the like.

(15) The substrate processing apparatus **100** includes a substantially cylindrical chamber **1** that is hermetically configured, and a susceptor **2** configured to horizontally support a substrate W, which is a substrate, is disposed in the chamber **1** in a state of being supported by a cylindrical support member **3** reaching the lower center of the susceptor **2** from the bottom of an exhaust chamber **21** to be described later. The susceptor **2** is made of ceramic such as AlN. A heater **4** is embedded in the susceptor **2**, and a heater power supply **5** is connected to the heater **4**. Meanwhile, a thermocouple **6** is provided near the top surface of the susceptor **2**, and a signal of the thermocouple **6** is transmitted to the heater controller **7**. The heater controller **7** transmits a command to the heater power supply **5** in response to the signal of the thermocouple **6** to control the heating of the heater **4** to control the substrate W to a predetermined temperature. The susceptor **2** is provided with three substrate lifting pins (not illustrated) to be capable of protruding or sinking with respect to the surface of the susceptor **2**, and the substrate lifting pins are in a state of protruding from the surface of the susceptor **2** when the substrate W is transported. The susceptor **2** is configured to be raised and lowered by a lifting mechanism (not illustrated).

(16) A circular hole **1b** is formed in the ceiling wall **1a** of the chamber **1**, and the shower head **10** is fitted to protrude into the chamber **1** from the circular hole **1b**. The shower head **10** ejects various processing gases supplied from a processing gas supply mechanism **8** to be described later into the chamber **1**. Above the shower head **10**, a first introduction path **11** configured to introduce a film-forming raw material gas (WCl.sub.6 gas) and a purge gas (N.sub.2 gas) and a second introduction path **12** configured to introduce a reduction gas (H.sub.2 gas) and a purge gas (N.sub.2 gas) are provided.

(17) Inside the shower head **10**, spaces **13** and **14** are provided in two upper and lower stages. The first introduction path **11** is connected to the upper space **13**. A first gas ejection path **15** extends from this space **13** to the bottom surface of the shower head **10**. The second introduction path **12** is connected to the lower space **14**. A second gas ejection path **16** extends from this space **14** to the bottom surface of the shower head **10**. That is, the shower head **10** is configured such that the film-forming raw material gas (WCl.sub.6 gas) and the reduction gas (H.sub.2 gas) are independently ejected from the gas ejection paths **15** and **16**, respectively.

(18) The bottom wall of the chamber **1** is provided with an exhaust chamber **21**, which protrudes

downward. An exhaust pipe **22** is connected to the side surface of the exhaust chamber **21**, and an exhaust device **23** including a vacuum pump, a pressure control valve, and the like is connected to the exhaust pipe **22**. By operating the exhaust device **23**, the interior of the chamber **1** can be brought into a predetermined depressurized (vacuum) state.

(19) The sidewall of the chamber **1** is provided with a carry-in/out port **24** configured to perform carry-in/out of a substrate **W** and a gate valve **25** configured to open/close the carry-in/out port **24**. Further, the wall of the chamber **1** is provided with a heater **26** so that the temperature of the inner wall of the chamber **1** can be controlled during a film forming process.

(20) The processing gas supply mechanism **8** includes a film-forming raw material gas supply line **30**, a reduction gas supply line **50**, a first purge gas supply line **60**, and a second purge gas supply line **70**.

(21) The film-forming raw material gas supply line **30** extends from a WCl.sub.6 gas supply mechanism **31**, which is a source of the film-forming raw material gas (WCl.sub.6), and is connected to the first introduction path **11**.

(22) The WCl.sub.6 gas supply mechanism **31** includes a raw material container **32** configured to accommodate a solid raw material **40** (WCl.sub.6) which is a film-forming raw material. WCl.sub.6 is a solid at room temperature, and WCl.sub.6 which is tungsten chloride as a tungsten raw material is accommodated in the raw material container **32** as a solid. A heater **32a** is provided around the raw material container **32**, and the film-forming raw material in the raw material container **32** is heated to an appropriate temperature to sublimate WCl.sub.6. The tungsten chloride is not limited to WCl.sub.6, and, for example, WCl.sub.5, WCl.sub.4, or the like may be used.

(23) A carrier gas supply line **33** configured to supply N.sub.2 gas, which is a carrier gas, is connected to the raw material container **32**. The carrier gas supply line **33** extends from the N.sub.2 gas source **34**, which is the source of the carrier gas (N.sub.2), and is connected to the raw material container **32**. The carrier gas supply line **33** is provided with a valve **35**, a mass flow controller **36**, a valve **37**, and a coupler **32b**, which will be described later, in this order from the N.sub.2 gas source **34**. The mass flow controller **36** controls the flow rate of the N.sub.2 gas flowing through the carrier gas supply line **33**. The valves **35** and **37** switch between supplying and stopping of the N.sub.2 gas. The carrier gas is not limited to the N.sub.2 gas, and may be another inert gas such as Ar gas.

(24) The film-forming raw material gas supply line **30** configured to supply the film-forming raw material gas is connected to the raw material container **32**. The film-forming raw material gas supply line **30** is provided with a coupler **32c**, a valve **38**, a mass flow meter **39**, and a merging portion **30a**, which will be described later, in this order from the raw material container **32**. The valve **38** switches between supplying and stopping of the WCl.sub.6 gas. The film-forming raw material gas supply line **30** is provided with a heater **30b** configured to prevent condensation of the WCl.sub.6 gas. The WCl.sub.6 gas sublimated in the raw material container **32** is carried by the N.sub.2 gas as a carrier gas (carrier N.sub.2) and is supplied into the shower head **10** via the film-forming raw material gas supply line **30** and the first introduction path **11**.

(25) The raw material container **32** configured to accommodate the solid raw material **40** is configured to be detachably attached to the carrier gas supply line **33** and the film-forming raw material gas supply line **30** via the couplers **32b** and **32c**.

(26) The reduction gas supply line **50** extends from a H.sub.2 gas source **51**, which is the source of the reduction gas (H.sub.2) and is connected to the second introduction path **12**. The reduction gas supply line **50** is provided with a valve **52**, a mass flow controller **53**, a valve **54**, and a merging portion **50a**, which will be described later, in this order from the H.sub.2 gas source **51**. The mass flow controller **53** controls the flow rate of the H.sub.2 gas flowing through the reduction gas supply line **50**. The valves **52** and **54** switch between supplying and stopping of the H.sub.2 gas. The reduction gas is not limited to H.sub.2 gas, and SiH.sub.4 gas, B.sub.2H.sub.6 gas, and NH.sub.3 gas may also be used. Two or more of the H.sub.2 gas, the SiH.sub.4 gas, the

B.sub.2H.sub.6 gas, and the NH.sub.3 gas may be supplied. Furthermore, other reduction gases other than these gases, for example, PH.sub.3 gas and SiH.sub.2Cl.sub.2 gas, may be used.

(27) The first purge gas supply line **60** extends from a N.sub.2 gas source **61**, which is the source of the purge gas (N.sub.2), and is connected to the merging portion **30a** of the film-forming raw material gas supply line **30**. The first purge gas supply line **60** is provided with a valve **62**, a mass flow controller **63**, and a valve **64** in this order from the N.sub.2 gas source **61**. The mass flow controller **63** controls the flow rate of the N.sub.2 gas flowing through the first purge gas supply line **60**. The valves **62** and **64** switch between supplying and stopping of N.sub.2 gas during an ALD process. The purge gas is not limited to the N.sub.2 gas, and may be another inert gas such as Ar gas.

(28) The second purge gas supply line **70** extends from a N.sub.2 gas source **71**, which is the source of the purge gas (N.sub.2), and is connected to the merging portion **50a** of the reduction gas supply line **50**. The second purge gas supply line **70** is provided with a valve **72**, a mass flow controller **73**, and a valve **74** in this order from the N.sub.2 gas source **71**. The mass flow controller **73** controls the flow rate of the N.sub.2 gas flowing through the second purge gas supply line **70**. The valves **72** and **74** switch between supplying and stopping of N.sub.2 gas during an ALD process. The purge gas is not limited to the N.sub.2 gas, and may be another inert gas such as Ar gas.

(29) The controller **9** controls the operation of each part of the substrate processing apparatus **100**. The controller **9** includes a central processing unit (CPU), a read only memory (ROM), and a random access memory (RAM). The CPU executes a desired process according to a recipe stored in a storage area of, for example, the RAM. In the recipe, apparatus control information for process conditions is set. The control information may include, for example, gas flow rate, pressure, temperature, and process time. A program used by the recipe and the controller **9** may be stored in, for example, a hard disk or a semiconductor memory. In addition, for example, the recipe may be set at a predetermined position to be read in the state of being stored in a storage medium readable by a portable computer, such as a CD-ROM or a DVD.

(30) Next, the operation of forming a tungsten film using the substrate processing apparatus **100** according to the present embodiment will be described with reference to FIGS. **2** and **3**.

(31) <Film Formation Through CVD Process>

(32) FIG. **2** illustrates an example of a gas supply sequence when forming a tungsten film through a CVD process.

(33) First, a substrate **W** is carried into the substrate processing apparatus **100**. Specifically, the controller **9** opens the gate valve **25**. Subsequently, the substrate **W** is carried into the chamber **1** through the carry-in/out port **24** by a transport arm (not illustrated), and the substrate **W** is placed on the susceptor **2**. The susceptor **2** is heated to a predetermined temperature (e.g., 350 degrees C. to 550 degrees C.) by the heater **4**. When the transport arm retracts from the carry-in/out port **24**, the gate valve **25** is closed. As the substrate **W**, for example, a substrate **W** in which a barrier metal film (e.g., TiN film or TiSiN film) is formed as a base film on the surface of a thermal oxide film or the surface of an interlayer insulating film having a recess, such as a trench or hole, may be used. Since the tungsten film has poor adhesion to the thermal oxide film and the interlayer insulating film and causes an incubation time to be long, it is difficult to form a film on the thermal oxide film or the interlayer insulating film, but by using a TiN film or a TiSiN film as a base film, film formation becomes easy. However, the base film is not limited thereto. In the following description, assuming that valves **35**, **37**, **52**, **62**, and **72** are open, the opening/closing control of the valves **38**, **54**, **64**, **74** will be described.

(34) In step **S1**, the pressure in the chamber **1** is increased, and the temperature of the substrate **W** is stabilized. Specifically, the controller **9** closes the valve **38**, closes the valve **54**, opens the valve **64**, and opens the valve **74**. As a result, the purge N.sub.2 gas of the N.sub.2 gas sources **61** and **71** is supplied into the chamber **1** to increase the pressure in the chamber **1**, and the temperature of the substrate **W** on the susceptor **2** is stabilized.

(35) After the pressure in the chamber **1** reaches a predetermined pressure, CVD film formation is performed in step S2. Specifically, the controller **9** opens the valve **38** and opens the valve **54** while flowing the purge N.sub.2 gas. Here, the carrier N.sub.2 gas of the N.sub.2 gas source **34** is supplied to the raw material container **32**. The solid WCl.sub.6 in the raw material container **32** is heated by the heater **32a** and is sublimated to generate WCl.sub.6 gas. As a result, the WCl.sub.6 gas is supplied from the WCl.sub.6 gas supply mechanism **31** with the carrier N.sub.2 gas into the chamber **1**. In addition, the H.sub.2 gas of the H.sub.2 gas source **51** is supplied into the chamber **1**. As a result, the WCl.sub.6 gas, which is a film-forming raw material gas, reacts with the H.sub.2 gas, which is a reduction gas, on the surface of the substrate W heated by the heater **4**, and a tungsten film is formed. The thickness of the tungsten film is controlled by a film formation time. As a result, a tungsten film having a desired thickness is formed.

(36) The controller **9** obtains the flow rate of the film-forming raw material gas based on the flow rate of the carrier gas by the mass flow controller **36** and the flow rate of the film-forming raw material gas and the carrier gas by the mass flow meter **39**. In addition, the controller **9** controls the flow rate of the carrier gas by the mass flow controller **36** and the temperature of the solid raw material **40** by the heater **32a** to control the flow rate of the film-forming raw material gas supplied to the chamber **1**.

(37) In step S3, the interior of the chamber **1** is purged. Specifically, the controller **9** closes the valve **38** and closes the valve **54** while flowing the purge N.sub.2 gas. As a result, the supply of the WCl.sub.6 gas and the H.sub.2 gas into the chamber **1** is stopped. In addition, the purge N.sub.2 gas is supplied into the chamber **1** to purge the WCl.sub.6 gas, the H.sub.2 gas, and a reaction product in the chamber **1**.

(38) Thereafter, the substrate W is carried out of the substrate processing apparatus **100**. Specifically, the controller **9** opens the gate valve **25**. Subsequently, the substrate W subjected to film formation is carried out from the chamber **1** through the carry-in/out port **24** by the transport arm (not illustrated). When the transport arm retracts from the carry-in/out port **24**, the gate valve **25** is closed.

(39) As described above, the substrate processing apparatus **100** according to the present embodiment forms a tungsten film through a CVD process.

(40) <Film Formation through ALD Process>

(41) FIG. 3 illustrates an example of a gas supply sequence when forming a tungsten film through an ALD process.

(42) First, a substrate W is carried into the substrate processing apparatus **100**. Specifically, the controller **9** opens the gate valve **25**. Subsequently, the substrate W is carried into the chamber **1** through the carry-in/out port **24** by a transport arm (not illustrated), and the substrate W is placed on the susceptor **2**. The susceptor **2** is heated to a predetermined temperature (e.g., 350 degrees C. to 550 degrees C.) by the heater **4**. When the transport arm retracts from the carry-in/out port **24**, the gate valve **25** is closed. As the substrate W, for example, a substrate W in which a barrier metal film (e.g., TiN film or TiSiN film) is formed as a base film on the surface of a thermal oxide film or the surface of an interlayer insulating film having a recess, such as a trench or hole, is used. Since the tungsten film has poor adhesion to the thermal oxide film and the interlayer insulating film and also causes an incubation time to be long, it is difficult to form a film on the thermal oxide film or the interlayer insulating film, but by using a TiN film or a TiSiN film as a base film, film formation becomes easy. However, the base film is not limited thereto. In the following description, assuming that valves **35**, **37**, **52**, **62**, and **72** are open, the opening/closing control of the valves **38**, **54**, **64**, and **74** will be described.

(43) In step S11, the pressure in the chamber **1** is increased, and the temperature of the substrate W is stabilized. Specifically, the controller **9** closes the valve **38**, closes the valve **54**, opens the valve **64**, and opens the valve **74**. As a result, the purge N.sub.2 gas of the N.sub.2 gas sources **61** and **71** is supplied into the chamber **1** to increase the pressure in the chamber **1**, and the temperature of the

substrate W on the susceptor 2 is stabilized.

(44) Hereinafter, ALD film formation is performed by repeating steps S12 to S15 for a predetermined number of cycles.

(45) In step S12, WCl.sub.6 gas, which is a film-forming raw material gas, is supplied into the chamber 1. Specifically, the controller 9 opens the valve 38 and supplies the WCl.sub.6 gas with a carrier N.sub.2 gas from the WCl.sub.6 gas supply mechanism 31 into the chamber 1 while supplying purge N.sub.2 gas from the first purge gas supply line 60 in the state in which the valve 64 is open and while supplying purge N.sub.2 gas from the second purge gas supply line 70 in the state in which the valve 74 is open. The WCl.sub.6 gas supplied into the chamber 1 is adsorbed on the surface of the substrate W.

(46) The controller 9 obtains the flow rate of the film-forming raw material gas based on the flow rate of the carrier gas by the mass flow controller 36 and the flow rate of the film-forming raw material gas and the carrier gas by the mass flow meter 39. In addition, the controller 9 controls the flow rate of the carrier gas by the mass flow controller 36 and the temperature of the solid raw material 40 by the heater 32a to control the flow rate of the film-forming raw material gas supplied to the chamber 1 to be constant.

(47) In step S13, the excess WCl.sub.6 gas in the chamber 1 is purged. Specifically, the controller 9 closes the valve 38. As a result, the supply of the WCl.sub.6 gas into the chamber 1 is stopped. In addition, the purge N.sub.2 gas is supplied into the chamber 1 from the first purge gas supply line 60 and the second purge gas supply line 70 to purge the excess WCl.sub.6 gas in the chamber 1.

(48) In step S14, H.sub.2 gas, which is a reduction gas, is supplied into the chamber 1. Specifically, the controller 9 opens the valve 54 and supplies the H.sub.2 gas from the H.sub.2 gas source 51 into the chamber 1 while supplying purge N.sub.2 gas from the first purge gas supply line 60 in the state in which the valve 64 is open and while supplying purge N.sub.2 gas from the second purge gas supply line 70 in the state in which the valve 74 is open. The H.sub.2 gas supplied into the chamber 1 reacts with (reduces) the WCl.sub.6 adsorbed on the surface of the substrate W to form a tungsten film. The thickness of the tungsten film is controlled by a film formation time.

(49) In step S15, the excess H.sub.2 gas and a reaction product in the chamber 1 are purged. Specifically, the controller 9 closes the valve 54. As a result, the supply of H.sub.2 gas into the chamber 1 is stopped. In addition, purge N.sub.2 gas is supplied into the chamber 1 from the first purge gas supply line 60 and the second purge gas supply line 70 to purge the excess H.sub.2 gas and the reaction product in the chamber 1.

(50) Hereinafter, steps S12 to S15 are repeated for a predetermined number of cycles to form a tungsten film having a desired thickness.

(51) Thereafter, the substrate W is carried out of the substrate processing apparatus 100. Specifically, the controller 9 opens the gate valve 25. Subsequently, the substrate W in which the film-formation has been completed is carried out from the chamber 1 via the carry-in/out port 24 by the transport arm (not illustrated). When the transport arm retracts from the carry-in/out port 24, the gate valve 25 is closed.

(52) As described above, the substrate processing apparatus 100 according to the present embodiment forms a tungsten film through an ALD process.

(53) <Raw Material Container>

(54) Next, the raw material container 32 configured to accommodate a solid raw material 40 will be further described with reference to FIGS. 4A to 4D and FIGS. 5 to 7. FIGS. 4A to 4D are schematic views illustrating examples of the raw material container 32. FIGS. 5 to 7 are examples of graphs illustrating the relationship between the weight of the solid raw material 40 and the total surface area of the solid raw material 40.

(55) When the solid raw material 40 accommodated in the raw material container 32 is used until the remaining amount of the solid raw material 40 is equal to or less than a predetermined amount, the vaporization amount of a raw material gas decreases. Thus, the raw material container 32 is

replaced. It is required to improve the usage rate of the solid raw material **40** accommodated in the raw material container **32** (the usage rate = $1 - \text{remaining amount at the time of replacement} / \text{initial filling amount}$). Therefore, it is required to improve the usage rate of the solid raw material **40** by increasing the initial filling amount (volume, weight) of the solid raw material **40** accommodated in the raw material container **32**. In order to reduce the frequency of replacement of the raw material container **32**, it is required to increase the initial filling amount (volume, weight) of the solid raw material **40** accommodated in the raw material container **32**.

(56) A solid raw material **40** having an initial particle size r_a and an initial weight (an initial filling amount) M_a is accommodated in the raw material container **32A** illustrated in FIG. 4A. The relationship between the weight and the total surface area in this configuration is illustrated by the black circles in FIG. 5. The total surface area of the solid raw material **40** at the initial stage is S_a . When the solid raw material **40** is vaporized and the raw material gas is released, the weight is reduced and the particle size is also reduced. Thus, the total surface area is also reduced.

(57) FIG. 4B illustrates a raw material container **132** according to a reference example. In the raw material container **132** according to the reference example, the initial weight of the solid raw material **40** is simply increased (e.g., 10 times). That is, the raw material container **132** includes the solid raw material **40** having an initial particle size r_a and an initial weight M_b ($M_b \gg M_a$). The relationship between the weight and the total surface area in this configuration is illustrated by the white circles and broken lines in FIG. 5. The total surface area of the solid raw material **40** at the initial stage is S_b ($S_b \gg S_a$). For example, when the initial filling amount is increased 10 times, the initial total surface area is also increased 10 times.

(58) Since the solid raw material **40** vaporizes from the surface thereof, the vaporization rate of the raw material gas depends on the total surface area of the solid raw material **40**. Therefore, when the raw material gas is supplied to the chamber **1** using the raw material container **132** illustrated in FIG. 4B, the amount of change in the total surface area of the solid raw material **40** from the initial stage (the state in which a new raw material container is attached) to the time of replacement (the state in which the raw material container is replaced) is greater than the amount of change in the total surface area of the solid raw material **40** from the initial time to the time of replacement in the raw material container **32A** illustrated in FIG. 4A. Therefore, adjustment parameters (the flow rate of a carrier gas by the mass flow controller **36** and the temperature of a raw material by the heater **32a**) in the case in which the raw material gas is supplied to the chamber **1** using the raw material container **132** illustrated in FIG. 4B are required to be changed more greatly than those in the case in which the raw material gas is supplied to the chamber **1** using the raw material container **32A** illustrated in FIG. 4A. For these reason, there is a concern that the adjustment parameters of the mass flow controller **36** and the heater **32a** deviate from a predetermined range.

(59) FIG. 4C illustrates another raw material container **32B** according to the present embodiment. The other raw material container **32B** according to the present embodiment increases the initial weight and adjusts the particle size of the solid raw material **40**. The other raw material container **32B** according to the present embodiment accommodates a solid raw material **40** having an initial particle size r_b and an initial weight M_b ($M_b \gg M_a$). The relationship between the weight and the total surface area in this configuration is illustrated by the black square marks and the solid line in FIG. 5. Here, the initial particle size r_b is adjusted such that the initial total surface area is equal to the initial total surface area S_a in the raw material container **32A** (or falls within a predetermined allowable range). Here, the initial particle size r_b is set to be greater than the initial particle size r_a . This makes it possible to maintain the initial total surface area at S_a while increasing the initial filling amount of the solid raw material **40** in the raw material container **32B**.

(60) As a result, even when the raw material container **32B** is used, it is possible to adjust the supply amount of the raw material gas within the ranges of the adjustment parameters (the flow rate of the carrier gas by the mass flow controller **36** and the temperature of the raw material by the heater **32a**) in the raw material container **32A**.

(61) In addition, as illustrated in FIG. 6, by controlling the initial particle size of the solid raw material **40** filled in the raw material container **32**, it is possible to increase the initial weight while maintaining the initial total surface area S_a . For example, by setting the initial particle size to r_c ($r_c > r_a$), it is possible to set the initial weight to M_c ($M_c > M_a$) while maintaining the initial total surface area at S_a . For example, by setting the initial particle size to r_d ($r_d > r_c > r_a$), it is possible to set the initial weight to M_d ($M_d > M_c > M_a$) while maintaining the initial total surface area at S_a . That is, by controlling the initial particle size, it is possible to impart the same initial total surface area for an arbitrary filling amount.

(62) In addition, as illustrated in FIG. 7, by controlling the initial particle size r_e of the solid raw material **40** filled in the raw material container **32**, it is possible to reduce the adjustment amount of a control parameter while maintaining the initial weight M_a . The initial total surface area S_e ($S_e < S_a$) is set to satisfy the necessary and sufficient supply amount of the raw material gas from the initiation of use of the raw material container **32** to the termination of use of the raw material container **32**.

(63) That is, when the solid raw material **40** is vaporized and the raw material gas is released, the weight is reduced and the particle size is also reduced. Thus, the total surface area is also reduced. Therefore, the vaporized amount of the solid raw material **40** is also reduced. In this case, the controller **9** maintains the supply amount of the raw material gas supplied to the chamber **1** by adjusting a control parameter. Here, as the control parameter, for example, the flow rate of the carrier gas by the mass flow controller **36** is used. When the vaporized amount of the solid raw material **40** is reduced, the supply amount of the raw material gas supplied to the chamber **1** is maintained by increasing the flow rate of the carrier gas. In addition, as the control parameter, for example, the heating temperature of the solid raw material **40** by the heater **32a** is used. When the vaporized amount of the solid raw material **40** is reduced, the heating temperature of the solid raw material **40** is raised to maintain the supply amount of the raw material gas supplied to the chamber **1**. As control parameters, both the flow rate of the carrier gas and the heating temperature may be controlled.

(64) Here, the greater the amount of change in the total surface area of the solid raw material **40** from the initiation of use of the raw material container **32** to the termination of use of the raw material container **32**, the greater the adjustment amount of the control parameter (flow rate of the carrier gas, heating temperature). When the adjustment amount of the control parameter is increased, the adjustment amount may deviate from an appropriate adjustable range in the CVD process or the ALD process, and it may be difficult to sufficiently use up the solid raw material **40** in the raw material container **32**.

(65) In contrast, as illustrated in FIG. 7, by adjusting the initial particle size of the solid raw material **40**, it is possible to reduce the amount of change in the total surface area of the solid raw material **40** from the initiation of use of the raw material container **32** to the termination of use of the raw material container **32**. This makes it possible to reduce the adjustment amount of the control parameter, so that it is possible to prevent the adjustment amount of the control parameter from being deviated from the appropriate adjustable range in the CVD process or the ALD process. Therefore, it becomes easy to sufficiently use up the solid raw material **40** in the raw material container **32**.

(66) Returning back to FIG. 1, the determination device **200** is a device for determining the initial filling state (the initial weight and the initial particle size of the solid raw material **40**) of the raw material container **32** attached to the substrate processing apparatus **100**. An operator who replaces the raw material container **32** attaches the raw material container **32** filled with the solid raw material **40** having the initial weight and the initial particle size determined by the determination device **200** to the substrate processing apparatus **100**.

(67) The determination device **200** is connected to the controller **9** of the substrate processing apparatus **100** to be communicable with the controller **9**, and the operation record (history) of the

substrate processing apparatus **100** is input to the determination device **200**. The operation record also includes information on the initial filling state of the raw material container **32** attached to the substrate processing apparatus **100**. The input operation record of the substrate processing apparatus **100** is stored in a storage part (not illustrated) of the determination device **200**. In addition, in the storage part of the determination device **200**, for example, a table corresponding to graphs illustrated in FIGS. 5 to 7 is stored. In the table the particle size, weight, and total surface area of the solid raw material **40** are associated with each other.

(68) The determination device **200** determines the weight and particle size of the solid raw material **40** to be filled in the raw material container **32** to be attached next time, based on the operation record of the substrate processing apparatus **100** and the table.

(69) For example, the determination device **200** determines the initial total area of the next solid raw material **40** based on the operation record. For example, the determination device **200** reads the initial total area of the past (previous) solid raw material **40** from the operation record of the storage part and determines the initial total area of the next solid raw material **40**. In addition, for example, the determination device **200** reads the maximum value and the minimum value of the initial total area of the solid raw material **40** having an operation record and determines the initial total area of the next solid raw material **40** from the range of the maximum value and the minimum value. As a result, the adjustment amount of the control parameter may be set to be within the range in which the operation record is available. When information on the initial filling state of the raw material container **32** included in the operation record stored in the storage part (not illustrated) is the initial weight and the initial particle size of the solid raw material **40**, the initial total surface area is calculated based on the table.

(70) In addition, the determination device **200** determines the initial weight (the initial filling amount) of the next solid raw material **40**. For example, the remaining number of substrates **W** to be processed by the substrate processing apparatus **100** is acquired based on the operation record. For example, when the remaining number of substrates **W** to be processed in the substrate processing apparatus **100** is sufficiently large, the initial weight of the next solid raw material **40** may be increased compared to the initial weight of the past (previous) solid raw material **40**. This makes it possible to reduce the frequency of replacement of the raw material container **32**. In addition, by increasing the initial filling amount, it is possible to improve the usage rate of the solid raw material **40** (the usage rate = $1 - \text{remaining amount at the time of replacement} / \text{initial filling amount}$). Furthermore, for example, the initial weight of the next solid raw material **40** may be reduced compared to the initial weight of the past (previous) solid raw material **40** depending on the remaining number of the substrates **W** to be processed in the substrate processing apparatus **100**. As a result, when all the substrates **W** are processed, the remaining amount of the solid raw material **40** in the raw material container **32** may be reduced. Therefore, the usage rate of the solid raw material **40** can be improved.

(71) The determination device **200** determines the initial particle size of the solid raw material **40** based on the determined initial total area, initial weight, and table. By using the table in which the particle size, weight, and total surface area of the solid raw material **40** are associated with each other, it is possible to determine the initial particle size that satisfies the determined initial total area and initial weight.

(72) The determination device **200** has been described with reference to the case in which the initial total area and the initial weight are determined based on an operation record and the initial particle size is determined based on the table by way of an example, but the present disclosure is not limited thereto. The determination device **200** may determine any two of the initial particle size, the initial weight, and the initial total area based on the operation record, and may determine the other one based on the table.

(73) In the foregoing, a film forming method according to the present embodiment using a substrate processing system according to the present embodiment has been described. However, the present

disclosure is not limited to the above-described embodiment or the like, and can be variously modified and improved within the scope of the present disclosure described in the claims.

(74) Although the controller **9** and the determination device **200** of the substrate processing apparatus **100** have been described as being provided separately, the controller **9** may also function as the determination device **200**.

(75) In addition, an example for determining the initial particle size of the solid raw material **40** based on the initial total area, the initial weight, and the table has been described using the determination device **200**, but the present disclosure is not limited thereto. For example, the function of the determination device **200** may be imparted to a host that is at a higher level than the apparatus (the substrate processing apparatus **100**). In this case, the host may manage and determine the initial particle size of the solid raw material **40** to be filled in each raw material container **32** for a plurality of apparatuses (the substrate processing apparatus **100**).

(76) For example, solid raw materials **40** having different particle diameters may be mixed with each other as illustrated in FIG. **4D**. In addition, controlling the particle size may include a case in which the mixing ratio of solid raw materials **40** having different particle sizes is changed.

(77) Although the solid raw material **40** has been described as being a raw material gas for forming a tungsten film, the raw material gas is not limited thereto, and the film to be formed is not limited thereto. Furthermore, the process performed on a substrate **W** is not limited to the film formation. That is, the process may be widely applied to a substrate processing apparatus **100** that vaporizes a solid raw material and supplies the vaporized raw material to the chamber **1**.

(78) Although the single-wafer type substrate processing apparatus **100** has been described as an example, the present disclosure is not limited thereto. The present disclosure may be applied to the film formation of a tungsten film in a multi-wafer type processing apparatus. Furthermore, the present disclosure may be applied to the film formation of a tungsten film in a batch type substrate processing apparatus.

(79) This application claims priority based on Japanese Patent Application No. 2019-114040 filed on Jun. 19, 2019, the disclosure of which is incorporated herein in its entirety by reference.

EXPLANATION OF REFERENCE NUMERALS

(80) **W**: substrate, **100**: substrate processing apparatus, **1**: chamber (processing container), **2**: susceptor, **3**: support member, **4**: heater (heating part), **8**: processing gas supply mechanism, **9**: controller, **30**: film-forming raw material gas supply line (raw material gas supplier), **31**: WCl.sub.6 gas supply mechanism, **32**: raw material container, **30b**, **32a**: heater, **32b**, **32c**: coupler (attachment portion), **33**: carrier gas supply line, **34**: N.sub.2 gas source, **36**: mass flow controller, **39**: mass flow meter, **50**: reduction gas supply line, **60**: first purge gas supply line, **70**: second purge gas supply line, **35**, **37**, **38**, **52**, **54**, **62**, **64**, **72**, **74**: valve, **53**, **63**, **73**: mass flow controller, **200**: determination device (determination portion)

Claims

1. A processing method performed by a substrate processing system, the substrate processing system including: a processing container including a stage on which a substrate is placed; a coupler to which a raw material container configured to accommodate a solid raw material is detachably attached; a heater configured to produce a raw material gas by vaporizing the solid raw material accommodated in the raw material container; a carrier gas supplier configured to supply a carrier gas to the raw material container; a supply line configured to supply the raw material gas with the carrier gas from the raw material container to the processing container; a mass flow controller configured to control a flow rate of the carrier gas supplied from the carrier gas supplier; a mass flow meter configured to measure a flow rate of the mixture of the raw material gas and the carrier gas; a memory configured to store: a table including information on a particle size, a weight, and a total surface area of the solid raw material which are associated with one another; and an operation

record including information on at least one initial total surface area of at least one past solid raw material, at least one initial weight of the at least one past solid raw material, at least one initial particle size of the at least one past solid raw material, and a remaining number of substrates to be processed by the substrate processing system; and a controller configured to: control at least one of the heater or the mass flow controller in order to control a flow rate of the raw material gas which is supplied to the processing container; the processing method comprising: determining, by the controller, an initial total surface area of a next solid raw material based on the at least one initial total surface area of the at least one past solid raw material based on the operation record, such that an adjustment amount of control parameters to control the heater and the mass flow controller, which is required to maintain a flow rate of a next raw material gas to the processing chamber to be constant, falls within a predetermined range, wherein the control parameters include the flow rate of the carrier gas through the supply line and a heating temperature of the next solid raw material; determining, by the controller, an initial weight of the next solid raw material based on the remaining number of substrates to be processed by the substrate processing system; determining, by the controller, an initial particle size of the next solid raw material based on the determined initial total surface area, the determined initial weight and the table, upon attachment of a next raw material container filled with the next solid raw material having the determined initial total surface area, the determined initial weight, and the determined initial particle size, determining, by the controller, the control parameters based on the determined initial total surface area of the next solid raw material; supplying, by the heater and the mass flow controller controlled by the controller, the next raw material gas from the next raw material container to the processing container based on the determined control parameters; obtaining, by the controller, the flow rate of the next raw material gas based on the flow rate of the carrier gas by the mass flow controller and a flow rate of the mixture of the next raw material gas and the carrier gas by the mass flow meter; controlling, by the controller, the heater and the mass flow controller to adjust the control parameters to maintain the flow rate of the next raw material gas to the processing chamber to be constant; and supplying, by the heater and the mass flow controller controlled by the controller, the next raw material gas based on the adjusted control parameters.

2. The processing method of claim 1, wherein an initial total surface area of the past solid raw material and an initial total surface area of the next solid raw material are equal to each other, and the initial weight of the next solid raw material is greater than an initial weight of the past solid raw material.

3. The processing method of claim 2, further comprising: determining the initial particle size of the next solid raw material such that the initial total surface area of the next solid raw material is maintained at a constant value of the initial total surface area of the past solid raw material and the initial weight of the next solid raw material becomes greater than the initial weight of the past solid raw material.

4. The processing method of claim 1, wherein an initial weight of the past solid raw material and the initial weight of the next solid raw material are equal to each other, and an amount of a change in a total surface area from initiation of use of the next raw material container to termination of use of the next raw material container is smaller than an amount of a change in a total surface area from initiation of use of the raw material container to termination of use of the raw material container.

5. The processing method of claim 4, further comprising: determining the initial particle size of the next solid raw material such that the initial weight of the next solid raw material is maintained at the initial weight of the past solid raw material and an initial total surface area of the next solid raw material becomes smaller than an initial total surface area of the past solid raw material.

6. The processing method of claim 1, further comprising: implementing a process on the substrate in a state in which the next raw material container is attached to the coupler.

7. A substrate processing system comprising: a processing container including a stage on which a substrate is placed; a coupler to which a raw material container configured to accommodate a solid

raw material is detachably attached; a heater configured to produce a raw material gas by vaporizing the solid raw material accommodated in the raw material container; a carrier gas supplier configured to supply a carrier gas to the raw material container; a supply line configured to supply the raw material gas with the carrier gas from the raw material container to the processing container; a mass flow controller configured to control a flow rate of the carrier gas supplied from the carrier gas supplier; a mass flow meter configured to measure a flow rate of the mixture of the raw material gas and the carrier gas; a memory configured to store: a table including information on a particle size, a weight, and a total surface area of the solid raw material which are associated with one another; and an operation record including information on at least one initial total surface area of at least one past solid raw material, at least one initial weight of the at least one past solid raw material, at least one initial particle size of the at least one past solid raw material, and a remaining number of substrates to be processed by the substrate processing system; and a controller configured to: control at least one of the heater or the mass flow controller in order to control a flow rate of the raw material gas which is supplied to the processing container; determine an initial total surface area of a next solid raw material based on the at least one initial total surface area of the at least one past solid raw material based on the operation record, such that an adjustment amount of control parameters to control the heater and the mass flow controller, which is required to maintain a flow rate of a next raw material gas to the processing chamber to be constant, falls within a predetermined range, wherein the control parameters include the flow rate of the carrier gas through the supply line and a heating temperature of the next solid raw material; determine an initial weight of the next solid raw material based on the remaining number of substrates to be processed by the substrate processing system; determine an initial particle size of the next solid raw material based on the determined initial total surface area, the determined initial weight, and the table, upon attachment of a next raw material container filled with the next solid raw material having the determined initial total surface area, the determined initial weight, and the determined initial particle size, determine the control parameters based on the determined initial total surface area of the next solid raw material; supply the next raw material gas from the next raw material container to the processing container based on the determined control parameters; obtain the flow rate of the next raw material gas based on the flow rate of the carrier gas by the mass flow controller and a flow rate of the mixture of the next raw material gas and the carrier gas by the mass flow meter; adjust the control parameters to maintain the flow rate of the next raw material gas to the processing chamber to be constant; and supply the next raw material gas based on the adjusted control parameters.

8. The substrate processing system of claim 7, wherein the controller is further configured to control at least one of the mass flow controller or the heater such that a supply amount of the raw material gas is constant.

9. The substrate processing system of claim 7, wherein an initial particle size of the past solid raw material and the initial particle size of the next solid raw material are different from each other.
